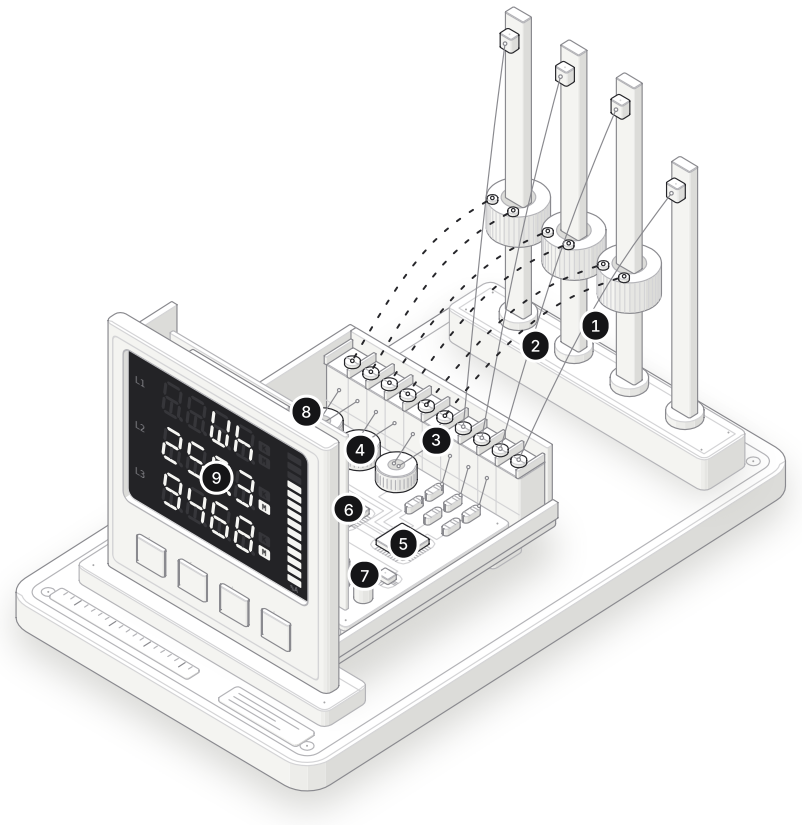
Energy meter: from bus bar to units

EM-001



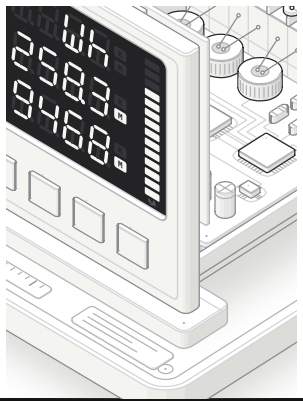
PARTS

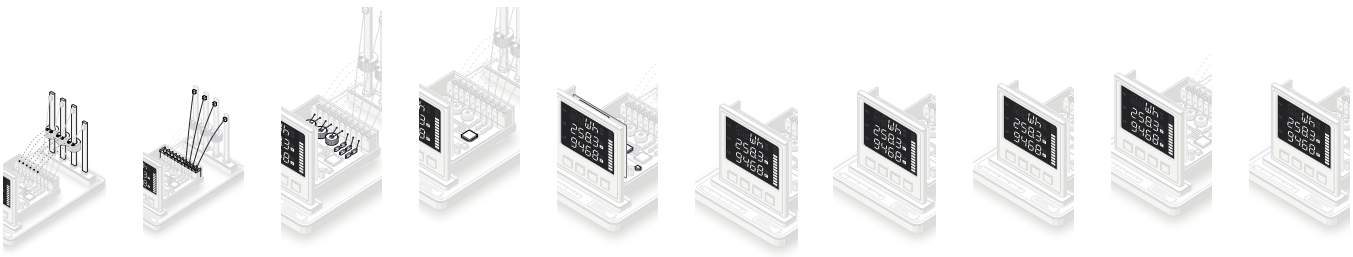
01 Bus bars and taps
02 External CTs 400 : 5
03 Terminal block
04 Internal CTs, divider
05 Metering IC
06 MCU, crystal, EEPROM
07 Power supply
08 Display board
09 Window and keys

SPECIFICATION

Supply	3 phase, 4 wire, 230 V L–N
CTs	400 : 5 on the bars
Power factor	0.9 (figure's model)
Display	3 rows × 4 digits, K and M lamps
Range	K up to 9999.9999 kWh, then M (figure's rule)

THE WINDOW





01 CTs step down
02 Voltage tapped
03 Scaled for the chip
04 Power is worked out
05 Energy is counted
06 1 · Top row is Wh
07 2 · Read row 2
08 3 · Add row 3
09 4 · K or M lamp
10 5 · Convert to units